Cancer

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भारतीय प्रौद्योगिकी संस्थान हैदराबाद

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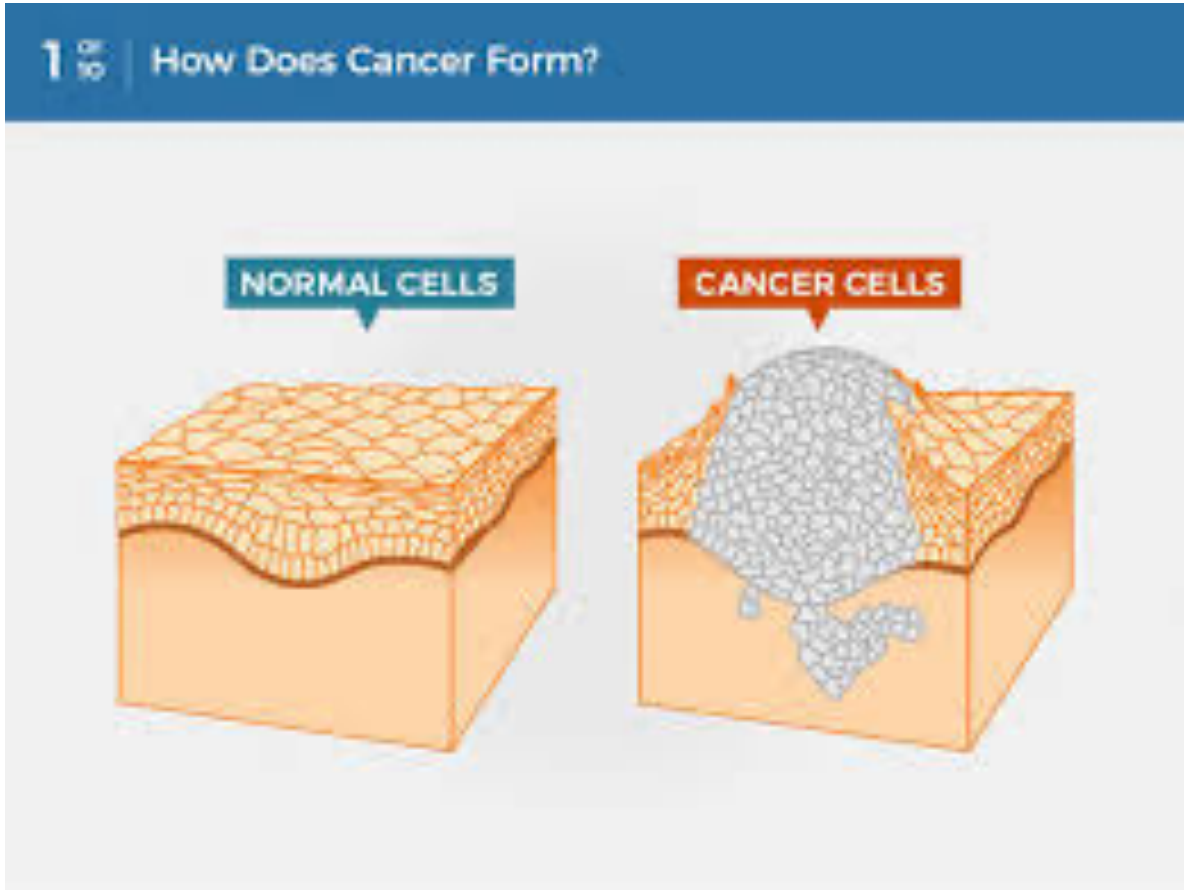
Cancer:

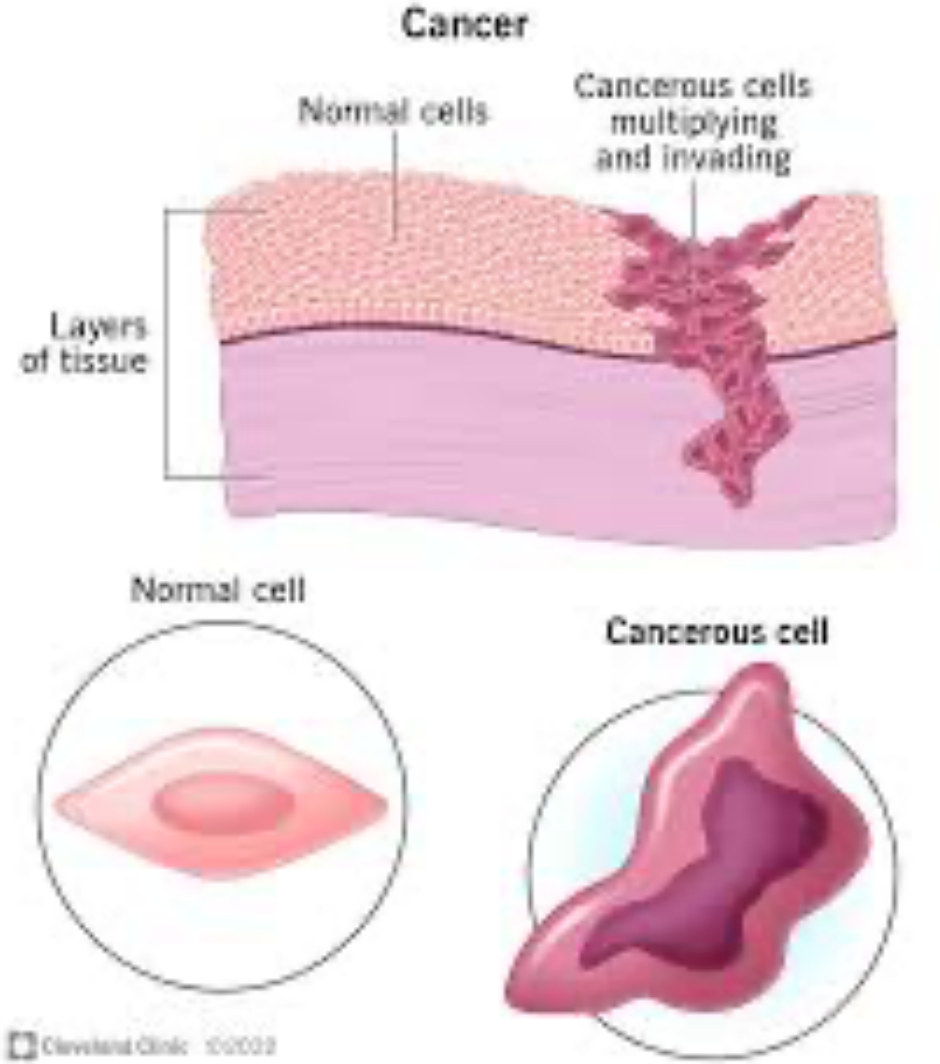
- Cancer is a group of diseases that cause abnormal cells to grow rapidly and spread to other parts of the body

Process of spreading:

- Cancer cells can break away from the original tumor and form new tumors in other parts of the body, a process called metastasis.

- Widespread metastases are the primary cause of death from cancer.


1 OF 10
How Does Cancer Form?
NORMAL CELLS
CANCER CELLS


Cancer
Normal cells
Cancerous cells multiplying and invading
Layers of tissue
Normal cell
Cancerous cell
Cleveland Clinic ©2022

- Most cancers form Tumor, but not all tumors are cancerous.
- Cancer is also called as Malignancy

Symptoms of Cancer:

Fatigue
Weight loss
Pain
Skin changes
Unusual bleeding
Persistent cough
Fever

Causative agents:

Chemical,
Toxic compound exposure,
Ionizing radiation,
Some pathogen

Treatment:

- Chemotherapy
- Radiation
- Surgery

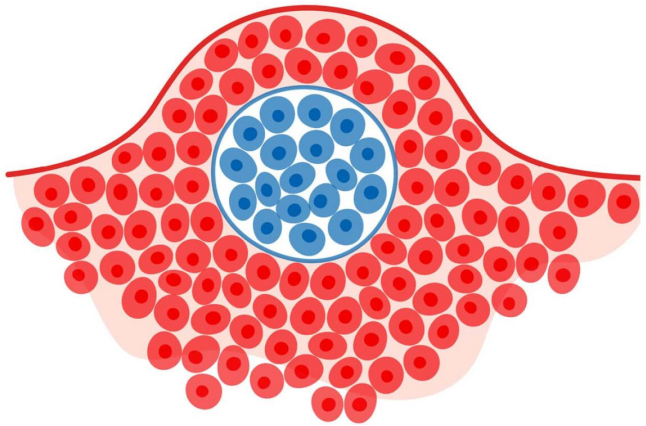
General Terms:

- **Benign:** non-cancerous, does not spread to other part of body.
- Ex: Uterine Fibroids, Lipomas, Meningiomas etc.

- **Malignant:** Cancerous, spread to other part of the body.
- Ex: Breast, lungs, skin, lungs, liver, ovaries, bone, prostate, brain,

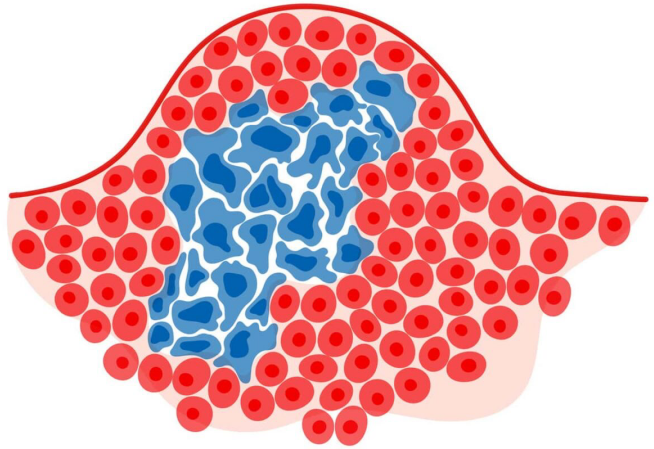
TUMOR TYPES

BENIGN TUMOR



- Non-cancerous.
- Capsulated.
- Non-invasive.
- Slow growing.
- Does not metastasize (spread) to other parts of the body.
- Cells are normal.

MALIGNANT TUMOR

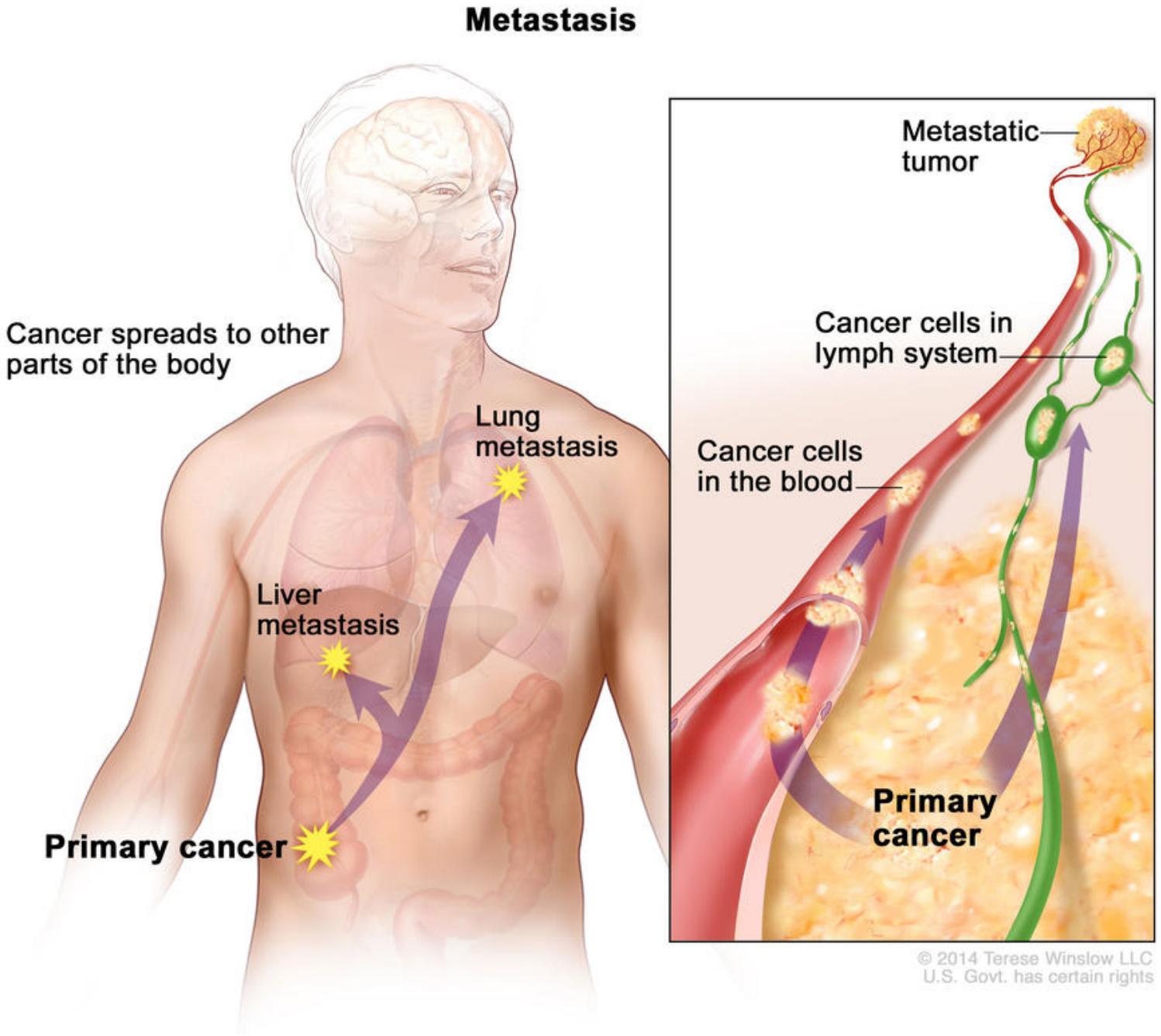


- Cancerous.
- Non-capsulated.
- Fast growing.
- Metastasizes (spreads) to other parts of the body.
- Cells have large, dark nuclei. May have abnormal shape.

- *Cancer that spreads from where it started to a distant part of the body is called metastatic cancer.*

- *For many types of cancer, it is also called stage IV (4) cancer.*

- *The process by which cancer cells spread to other parts of the body is called metastasis.*



- When observed under a microscope and tested in other ways, metastatic cancer cells have features like that of the primary cancer and not like the cells in the place where the metastatic cancer is found.
- This is how doctors can tell that it is cancer that has spread from another part of the body.

- Metastatic cancer has the same name as the primary cancer.
- For example, breast cancer that spreads to the lung is called metastatic breast cancer, not lung cancer. It is treated as stage IV breast cancer, not as lung cancer.

- Sometimes when people are diagnosed with metastatic cancer, doctors cannot tell where it started. This type of cancer is called cancer of unknown primary origin, or (CUP) Carcinoma of Unknown Primary.

How Cancer Spreads

Cancer cells spread through the body in a series of steps. These steps include:

- *Growing into, or invading, nearby normal tissue*
- *Moving through the walls of nearby lymph nodes or blood vessels*
- *Traveling through the lymphatic system and bloodstream to other parts of the body*
- *Stopping in small blood vessels at a distant location, invading the blood vessel walls, and moving into the surrounding tissue*
- *Growing in this tissue until a tiny tumor forms*

causing new blood vessels to grow, which creates a blood supply that allows the metastatic tumor to continue growing

Most of the time, spreading cancer cells die at some point in this process.
But, as long as conditions are favorable for the cancer cells at every step, some of them are able to form new tumors in other parts of the body.
Metastatic cancer cells can also remain inactive at a distant site for many years before they begin to grow again, if at all.

Common Sites Where Cancer Spreads

Cancer Type	Main Sites of Metastasis
Ovary	Liver, lung, peritoneum
Pancreas	Liver, lung, peritoneum
Prostate	Adrenal gland, bone, liver, lung
Rectal	Liver, lung, peritoneum
Stomach	Liver, lung, peritoneum
Thyroid	Bone, liver, lung
Uterus	Bone, liver, lung, peritoneum, vagina

Cancer Type	Main Sites of Metastasis
Bladder	Bone, liver, lung
Breast	Bone, brain, liver, lung
Colon	Liver, lung, peritoneum
Kidney	Adrenal gland, bone, brain, liver, lung
Lung	Adrenal gland, bone, brain, liver, other lung
Melanoma	Bone, brain, liver, lung, skin, muscle

Symptoms of Metastatic Cancer

- Metastatic cancer does not always cause symptoms.
- When symptoms do occur, what they are like and how often you have them will depend on the size and location of the metastatic tumors.

Some common signs of metastatic cancer include:

- Pain and fractures, when cancer has spread to the bone
- Headache, seizures, or dizziness, when cancer has spread to the brain
- Shortness of breath, when cancer has spread to the lung
- Jaundice or swelling in the belly, when cancer has spread to the liver

Treatment for Metastatic Cancer

- There are treatments for most types of metastatic cancer.
- Often, the goal of treating metastatic cancer is to control it by stopping or slowing its growth.
- Some people can live for years with metastatic cancer that is well controlled. Other treatments may improve the quality of life by relieving symptoms.

Ongoing Research

Researchers are studying new ways to kill or stop the growth of primary and metastatic cancer cells. These ways include:

- Helping your immune system fight cancer
- Disrupting the steps in the process that allow the cancer cells to spread
- Targeting specific genetic changes in tumors

Carcinoma:

Cancer that begins in the skin or in tissues that line or cover internal organs.

Sarcoma:

A type of cancer that begins in bone or in the soft tissues of the body, including cartilage, fat, muscle, blood vessels, fibrous tissue, or other connective or supportive tissue.

Leukemia:

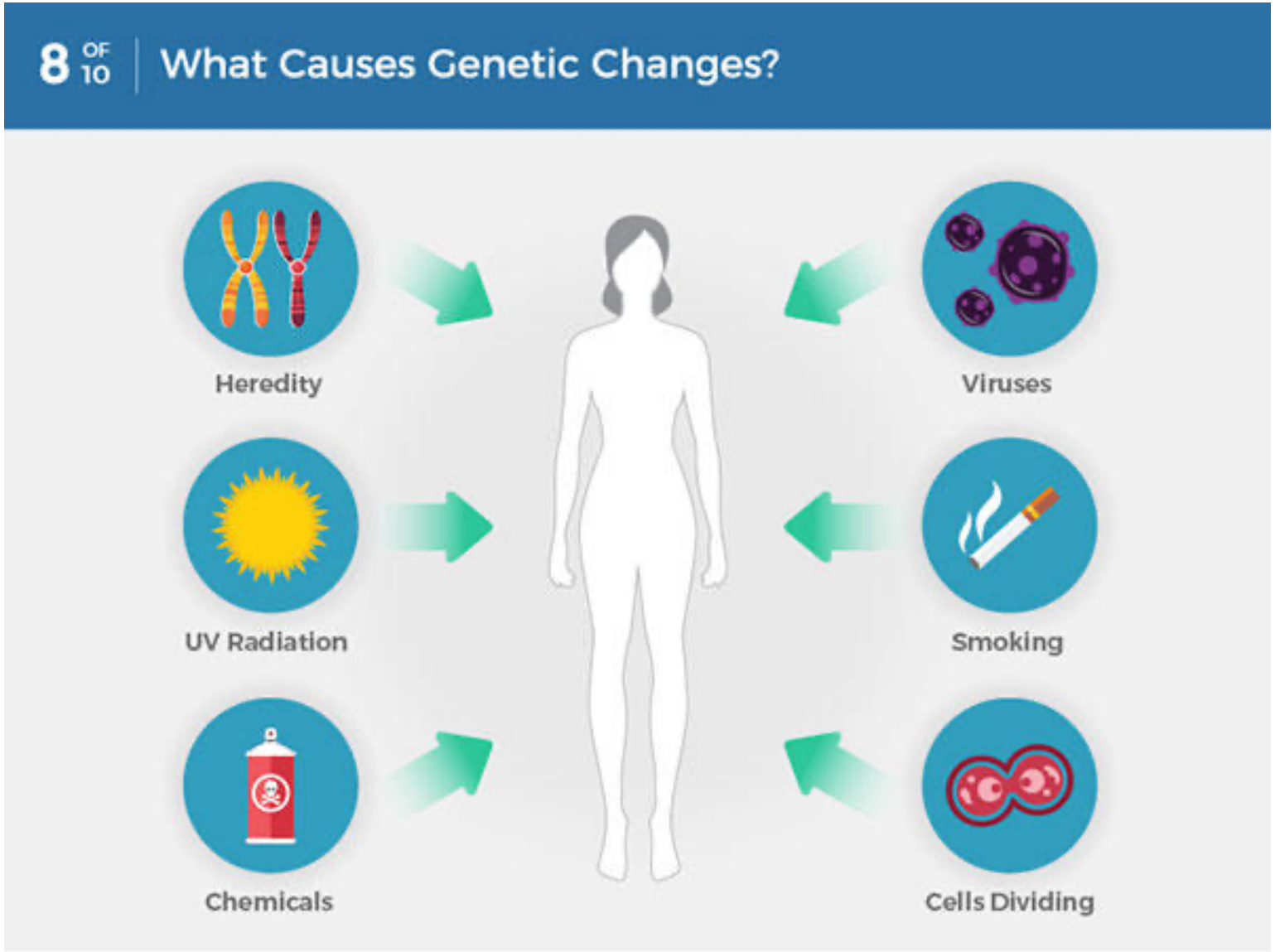
Cancer that starts in blood-forming tissue, such as the bone marrow, and causes large numbers of abnormal blood cells to be produced and enter the bloodstream.

The differences between normal cells and cancer cells

Normal cells	Cancer cells
Controlled growth	Uncontrolled growth
Normal chromosomes	Abnormal chromosomes
Undergo apoptosis	No apoptosis
Contact inhibition	No contact inhibition
Stay in organ	Ability to metastasize
Specialized cells	Nonspecialized cells
Undergo maturation and senescence	Immature/undifferentiated
Even appearance	Variable appearance
Even shaped nuclei	Variable shaped nuclei
Organized arrangement	Disorganized arrangement

Is cancer a genetic disease?

- Yes, cancer is a genetic disease.
- It is caused by changes in genes that control the way cells grow and multiply.
- DNA changes, whether caused by a random mistake or by a carcinogen, can happen throughout our lives and even in the womb.



Is cancer hereditary?

- Cancer itself can't be passed down from parents to children.

And genetic changes in tumor cells can't be passed down.

- But a genetic change that increases the risk of cancer can be passed down (inherited) if it is present in a parent's egg or sperm cells.

For example, if a parent passes a mutated BRCA1 or BRCA2 gene to their child, the child will have a much higher risk of developing breast and several other cancers.

That's why cancer sometimes appears to run in families. Up to 10% of all cancers may be caused by inherited genetic changes.

- *That's why cancer sometimes appears to run in families. Up to 10% of all cancers may be caused by inherited genetic changes.*
- *Inheriting a cancer-related genetic change doesn't mean you will definitely get cancer. It means that your risk of getting cancer is increased.*

Should one get genetic testing for cancer risk?

- *Certain genetic tests can show if you've inherited a genetic change that increases your risk of cancer.*
- *This testing is usually done with a small sample of blood, but it can sometimes be done with saliva, cells from inside the cheek, or skin cells.*



Genetic tests can help families with a history of breast and ovarian cancer make screening and treatment decisions

AIDS

Acquired Immunodeficiency Syndrome

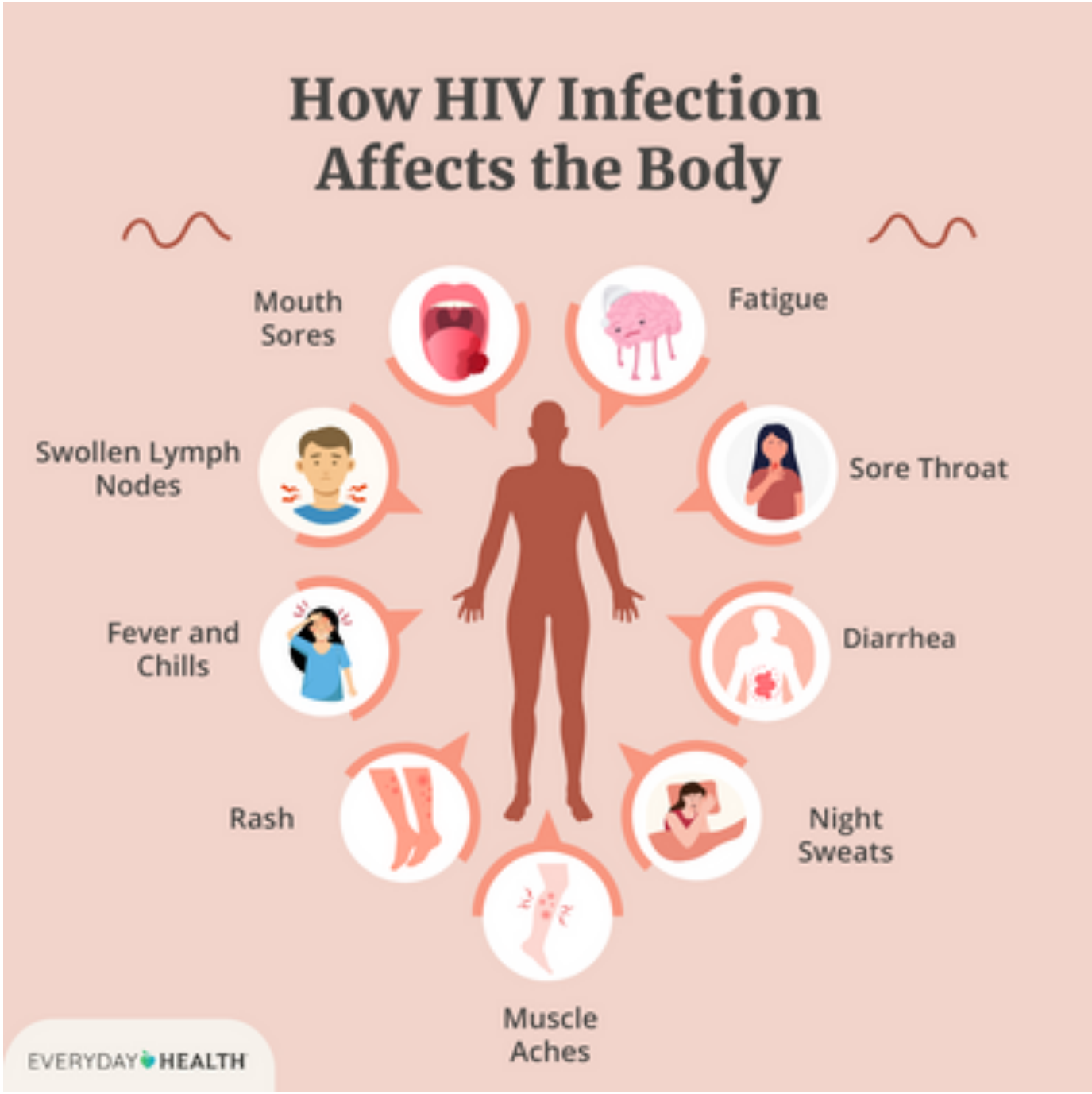
- A disease caused by the human immunodeficiency virus (HIV).

What's the Difference Between HIV and AIDS?

- HIV stands for Human Immunodeficiency Virus.

It's a virus that attacks the body's immune system, weakening your defenses against infections.

- AIDS, on the other hand, stands for Acquired Immunodeficiency Syndrome.
- It's the most severe stage of HIV infection, where the immune system is so damaged that it can no longer fight off certain infections or cancers.


How HIV Infection Affects the Body
Mouth Sores
Fatigue
Swollen Lymph Nodes
Sore Throat
Fever and Chills
Diarrhea
Rash
Night Sweats
Muscle Aches
EVERYDAY HEALTH

HIV/AIDS is transmitted



UNPROTECTED SEX

DRUG ADDICTS

BLOOD TRANSFUSION

PREGNANCY

NON-STERILE INSTRUMENTS

HIV/AIDS is not transmitted



TOUCHING

THROUGH FOOD

WITH A KISS

INSECT BITES

IN THE POOL

HIV Diagnosis

- Blood Test
- ELISA (Enzyme Linked Immunosorbent Assay)

